

Problem Statement :

Predict the insurance charges based on the given parameters.

Domain Selection :

Machine Learning is the domain as the input set is numerical values.

Learning Method :

Supervised Learning , Regression : Requirement is clear and dataset has input and output, output is numerical value.

Dataset info:

Inputs :

age	sex	bmi	children	smoker
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Number of records : 1338

Preprocessing Method:

Input set has two columns (sex , smoker) which contains nominal values, for converting nominal value in to numerical value we have to use one hot encoding which involves column expansion.

Model Creation:

Best model can be found only by applying the dataset in all algorithms.

Based on the R^2 value of each model we can find the best model .

1. Multiple Linear Regression:

R^2 value = 0.7979

2. Support Vector Machine:

SL. No	Hyper Parameter	Kernel = Linear	Kernel = rbf	Kernel = poly	Kernal = sigmoid
1	C=10	0.03355	-0.0424	0.0459	0.0427
2	C=100	0.5662	0.3547	0.6591	0.5353
3	C=1000	0.6666	0.8283	0.8631	0.172
4	C=10000	0.7495	0.8867	0.8619	-42.53

R² value = 0.8867

3. Decision Tree:

SL No	Criterion	Splitter	Max Features	R ² Value
1	squared_error	best	None	0.7733
2	friedman_mse	best	None	0.7842
3	absolute_error	best	None	0.7346
4	poisson	best	None	0.7838
5	squared_error	random	None	0.7211
6	friedman_mse	random	None	0.7443
7	absolute_error	random	None	0.7746
8	poisson	random	None	0.6969
9	squared_error	best	100	0.7609
10	friedman_mse	best	100	0.7726
11	absolute_error	best	100	0.7392
12	poisson	best	100	0.7771
13	squared_error	best	0.9	0.7108
14	friedman_mse	best	0.9	0.7238
15	absolute_error	best	0.9	0.7801
16	poisson	best	0.9	0.7132
17	squared_error	best	sqrt	0.784
18	friedman_mse	best	sqrt	0.7579
19	absolute_error	best	sqrt	0.739
20	poisson	best	sqrt	0.6823
21	squared_error	best	log2	0.7809
22	friedman_mse	best	log2	0.8011
23	absolute_error	best	log2	0.8101
24	poisson	best	log2	0.5592
25	squared_error	random	100	0.7405
26	friedman_mse	random	100	0.7135
27	absolute_error	random	100	0.7508
28	poisson	random	100	0.7108
29	squared_error	random	0.9	0.7451
30	friedman_mse	random	0.9	0.8179
31	absolute_error	random	0.9	0.7992
32	poisson	random	0.9	0.7156
33	squared_error	random	sqrt	0.6546
34	friedman_mse	random	sqrt	0.6662
35	absolute_error	random	sqrt	0.6542
36	poisson	random	sqrt	0.7352

37	squared_error	random	log2	0.6989
38	friedman_mse	random	log2	0.7616
39	absolute_error	random	log2	0.6852
40	poisson	random	log2	0.7667

R² value = 0.8179

4. Random Forest:

SL No	Criterion	Max Features	n_estimators	R2
1	squared_error	0.3	100	0.8739
2	squared_error	log2	100	0.8939
3	friedman_mse	0.3	100	0.8739
4	friedman_mse	log2	100	0.8938
5	absolute_error	0.3	100	0.8625
6	absolute_error	log2	100	0.8891
8	poisson	0.3	100	0.8713
9	poisson	log2	100	0.8938
10	squared_error	0.3	50	0.8709
11	squared_error	log2	50	0.8924
12	friedman_mse	0.3	50	0.8709
13	friedman_mse	log2	50	0.8922
14	absolute_error	0.3	50	0.865
15	absolute_error	log2	50	0.8885
16	poisson	0.3	50	0.863
17	poisson	log2	50	0.8916

R² value = 0.8939

Final Machine Learning BEST method of regression is

Random Forest R² value (max_features= 'log2'/sqrt',n_estimators= 100)= 0.8939